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AIRs - LM in

Business Mathematics

Week 9: Module 13

GOVERNMENT PROPERTY
NOT FOR SALE

Business Mathematics

Week 9: Module 13

Target

Typically a mortgage is obtained for a home, a house and a lot, or for a machinery or equipment. A loan is a debt device that the borrower is obliged to pay lower back with a predetermined set of installments, sponsored through collateral of said immovable property. Today, the Internet has what is referred to as a mortgage amortization calculator which helps prospective homeowners. They only enter the place, the amount of the loan, the duration, and the APR (Annual Percentage Rate), and it will provide the sum of their monthly payment, the average of the payments for the whole loan period, and the repayment date or the date they will have to pay the loan in full.

The learning competency in the previous lesson is solve problems involving buying and selling products.

After going through this module, you are expected to:

1. Compute interest specifically as applied to mortgage, amortization, and on services/utilities and on deposits and loans. **(ABM_BM11BS-Ij-10)**

Before going on, check how much you know about this topic. Answer the pretest on the next page in a separate sheet of paper.

Module 2	Mortgages

Mortgage will be discussed in this module since the previous competency is solving problems about buying and selling product. In this module, you will learn how to compute for the down payment on a mortgage and the amount of the mortgage loan.

PRETEST

Directions: Choose your answer from the given choices.

1. What do you call the loan uses a property as a collateral for a loan from a financial institution?
   A. Installment payment B. Down payment C. Mortgage loan

2. What do you call the certain percent of the purchase price of the property?
   A. Installment payment B. Down payment C. Mortgage loan

3. What do you call the agreement or contract involving a loan that is repaid over time with a set number of scheduled payments?
   A. Installment payment B. Down payment C. Mortgage loan

4. You are planning to buy a house and lot and secure a mortgage given that the price of the house and lot is P2,500,000.00. The real estate dealer is asking for a 25% down payment. Compute the down payment.
   A. P625,000 B. P625,500 C. P626,000

5. You are planning to buy a house and lot and secure a mortgage given that the price of the house and lot is P2,500,000.00. The real estate dealer is asking for a 25% down payment. How much is the mortgage loan?
   A. P1,865,000 B. P1,875,000 C. P1,885,000

Jumpstart

For you to understand the lesson well, do the following activities. Have fun and good luck!

Activity 1: Read Me! Solve ME!

You are planning to buy a house and lot and secure a mortgage given that the price of the house and lot is P3,000,000.00. The real estate dealer is asking for a 25% down payment:

a. How much is the down payment?
b. How much would be the amount of the mortgage loan?
c. Assume that you are to pay P13,489.70 monthly for 30 years at an interest rate of 6%.
d. How much would go to the interest and how much to the principal?

Read the following about mortgage and how to compute down payment and mortgage loan.

Mortgage loan is when a financial institution uses the home as collateral for a loan. A down payment is usually expected for most installment purchases. This typically amounts to several percent of the purchase price. A mortgage loan is a long-term loan. Usually, a borrower will choose a loan term between 5 and 30 years. Many institutions offer a 50-year term loan, but the longer it takes to pay off a mortgage, the higher the interest rate.

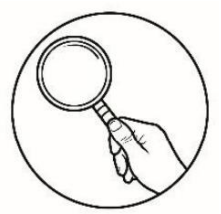
A property mortgage requires payment of the purchase price for the property at the most basic level, and interest on the loan. The down payment is usually some fraction of the property's purchase price. It is usually called the equity of the buyer. Suppose you want to buy a secondhand car worth P310,000 and the seller wants a down payment of 20 percent.

To compute for the down payment:

$$\text{Down payment} = \text{Purchase price} \times \text{Down payment } \%$$
$$= \text{P}310{,}000 \times 20\%$$
$$= \text{P}62{,}000$$

Therefore, if you pay the 20% down payment, the amount of the mortgage loan would be the balance after the down payment has been deducted from the purchase price, which in our example would be:

Mortgage loan = Purchase price – Down payment
= P310,000 – P62,000
= 248,000



Discover

Types of Mortgages

The two most common types of mortgages are fixed-rate and adjustable-rate (also known as variable rate) mortgages.

Fixed-Rate Mortgages

Fixed-rate mortgages provide lenders with a fixed interest rate for a defined period of usually 15, 20, or 30 years. For a fixed interest rate, the shorter the period in which the creditor charges, the higher the monthly charge. Conversely, the longer it takes the borrower to pay, the smaller the monthly amount of repayment. However, the longer it takes for the credit to be repaid, the more the borrower ultimately pays in interest charges.

The biggest advantage of a fixed-rate mortgage is that the homeowner can count on their monthly mortgage payments being the same every month over their mortgage life, making it easier to set household expenses and avoid any unnecessary extra costs from one month to the next. However, if market prices grow sharply, the borrower will not have to make higher monthly payments.

Adjustable-Rate Mortgages

Adjustable mortgages (ARMs) come with interest rates that can change over the life of the loan, and usually do so. Market rate increases and other factors cause interest rates to fluctuate, which changes the amount of interest the borrower has to pay, and thus changes the total monthly payment due. The interest rate is to be assessed and adjusted once a year or once every six months for adjustable mortgages.

Term of the Loan: Total Number of Payments

The billing of the loan in installments is called amortization. The schedule prepared which shows the interest payments for the payment duration (called the loan term) is called the amortization table. Suppose you have to make one payment per month for 5 years in our example in Module 2. The loan term is five years. Five years multiplied by 12 months per year will give you 60, meaning you have to make 60 monthly payments to be able to fully pay your loan.

Monthly Payment

If there was no interest rate it would be easy to calculate your monthly rate. Divide a loan sum by payment number ($\frac{248,000}{60}$ = P4,133.33). The bank must make money, however, so that interest is collected by the bank. Suppose the bank charges 5 percent per annum. The 5% is what is called the annual percentage (APR) average. Since you are paying monthly, we have to convert the 5% APR to a monthly rate. We divide 5% by 12 to get 0.416% (5%/12 =0.416%). To determine the monthly payment, we use the following formula:

$$A = \frac{i \times P \times (1+i)^{\mathbf{n}}}{(1+i)^{\mathbf{n}} - 1}$$

where A is the monthly payment;
P is the loan's initial amount;
i is the monthly interest rate; and
n is the total number of payments.

Using our example,

$$A = \frac{(0.00416)(P248,000) \times (1+0.00416)^{\mathbf{60}}}{(1+0.00416)^{\mathbf{60}} - 1}$$

= P4,679.17

The amortization plan is a table or chart showing each monthly payment on an amortizing loan indicating how much of each payment goes to interest, and how much goes to principal. When the sum goes to interest reduces when payment is made, so raises the amount that goes to the principal. Generally speaking the amortization table refers to fixed mortgages/loans.

Using our example, a second-hand car with a 20 percent down payment is worth P310,000.00. Your bank has decided to give you a P248,000.00 mortgage for 5 years at a fixed interest rate of 5 percent. What is the monthly payment? How much money do you pay each month for interest and principal?

The loan sum is P248,000.00 which is the principal. When you pay for 60 months P4,679.17, it is going to equal P280,750.20. So if you subtract the amount of the loan from the overall payments (P280,750.20 – P248, 000), the amount of interest you would have to pay for the entire loan period will be P32,750.20.

Amortization Table

Although the monthly payment is constant, the amount of money charged to interest and principal varies on a monthly basis. For the first payment, we multiply the principal balance of P248,000 by the monthly interest rate of 0.00416 to get P1,031.68 to determine what portion of P4,679.17 goes to interest. Therefore, deducting P1, 031.68 from P4,679.17 payment, it will result in P3,647.49. Deducting the P3,647.49 from P248,000, the new balance of the mortgage loan will now be P244,352.51.

If we continue this process, we can have an amortization table created. The use of the Internet calculator for the amortization plan is typically resorted to. The following amortization plan refers to our example as set out in the myamortizationchart.com website:

Payment Summary

No.of Payments	Monthly Payment	Total Principal Paid	Total Interest Paid	Total paid
60	P4,680.07	P248,000	P32,603.96	P280,803.96
Monthly Amortization Schedule				
Payment	Amount	Principal	Interest	Balance
1	P4,680.07	P3,646.73	P1,033.33	P244,353.27
2	P4,680.07	P3,661.93	P1,018.14	P240,691.34
3	P4,680.07	P3,677.19	P1,002.88	P237,014.15
4	P4,680.07	P3,692.51	P987.56	P233,321.65
5	P4,680.07	P3,707.89	P972.17	P229,613.76

56	P4,680.07	P4,583.77	P96.29	P18,526.87
57	P4,680.07	P4,602.87	P77.20	P13,924.00
58	P4,680.07	P4,622.06	P58.02	P9,301.95
59	P4,680.07	P4,641.31	P38.76	P4,660.65
60	P4,680.07	P4,660.65	P19.42	P0.00

Gauge

Directions: Read carefully each item. Use a separate sheet for your answers. Write only the letter of the best answer for each test item.

1. What do you name a loan which is used as collateral for property or real estate?
   A. Educational loan
   B. Financial loan
   C. Mortgage
   D. Salary loan
2. What are the components of your monthly mortgage payment?
   A. principal, interest, escrow
   B. principal, interest escrow, taxes
   C. principal, interest, taxes, insurance
   D. principal, interest, taxes, amortization fees
3. What do you name the arrangement or contract involving a loan with a fixed number of scheduled payments that is repaid over time?
   A. Installment payment
   B. Interest
   C. Down payment
   D. Mortgage loan
4. What do you name that certain percentage of the property's purchase price?
   A. Down payment
   B. Installment payment
   C. Mortgage loan
   D. Interest
5. What do you name that certain percentage of the property's purchase price?
   A. Adjustable-rate mortgages
   B. Conventional mortgage
   C. Fixed-rate mortgage
   D. Home loan
6. If you make one extra payment a year on a 30-year fixed mortgage, by how much will it shorten the term of your loan?
   A. one year
   B. two years
   C. five years
   D. 10 years
7. How many years does to pay his/her loan?
   A. 20 years
   B. between 5 and 30 years
   C. 5 years to 30 years
   D. between 2 and 35 years
8. How many different terms (lengths) are available for a fixed-rate mortgage?
   A. One
   B. Two
   C. three
   D. five
9. What's the easiest term to qualify for: 15, 20 or 30 years?
   A. 15
   B. 20
   C. 30
   D. None of these
10. Which form of mortgage has interest rates that fluctuate over the entire life of the loan (depending on the market)?
    A. Adjustable-rate mortgages
    B. Conventional mortgage
    C. Fixed-rate mortgage
    D. Home loan
11. What does ARM stand for?
    A. adjustable-rate mortgage
    B. adjustable-reasonable mortgage
    C. annual return mortgage
    D. annual rate mortgage

12. The average rate for a 30 year fixed mortgage is 3.48%. Assume a down payment of 20% on a home purchase of P3,000,000. How much is the down payment?
    A. P6,000
    B. P60,000
    C. P600,000
    D. P610,000
13. The average rate for a 30 year fixed mortgage is 3.48%. Assume a down payment of 20% on a home purchase of P3,000,000. How much is the mortgage loan?
    A. P2,400,400
    B. P2,400,000
    C. P2,940,000
    D. P2,994,000
14. The average rate for a 30 year fixed mortgage is 3.48%. Assume a down payment of 20% on a home purchase of P3,000,000. How much is the monthly payment?
    A. P10,750.85
    B. P10,751.30
    C. P10,850.85
    D. P10,890.50
15. The average rate for a 30 year fixed mortgage is 3.48%. Assume a down payment of 20% on a home purchase of P3,000,000. How much you have to pay for the entire term of the loan?
    A. P1,470,300
    B. P1,470,306
    C. P1,570,306
    D. P1,740,630

Great job! You are almost done with this module.

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BUSINESS MATH QUARTER 3

Most Essential Learning Competencies	Output To Be Submitted	Date of Completion
Quarter III Week 1 ***2 Fractions/ decimals and percentage*** **February 21-25, 2022** ✓ Give real-life situations to illustrate fractions, decimals, and percent (ABM_BM11FO-Ic-4) ✓ Solve problems involving fractions, decimals and percent (ABM_BM11FO-Id-5)	**Worksheet for Week 1 -2 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 1-2 . Gauge. Activity 8. Convert Me! (A, B, C, D, E, F) o Write all your answers on yellow paper. o Write the activity title o Write the quarter number and week number on the top right side of the paper.	**March 3, 2022**
Quarter III Week 3 – ***Key Concepts of Ratio and Proportion*** **February 28 – March 4, 2022** ✓ Identify the different kinds of proportions and write examples of real-life situations for each	**Worksheet for Week 3 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 3. Explore. Activity 1. #s 1-5	**March 7, 2022**
Quarter III Week 4 ***– Key Concepts of Ratio and Proportion*** **March 7-11, 2022** ✓ Solve problems involving direct, inverse and partitive proportion (ABM_BM11RP-If-4)	**Worksheet for Week 4 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 4. Gauge #s 1-15	**March 14, 2022**
Quarter III Week 5 ***Key Concepts of Buying and Selling*** **March 14-18, 2022** ✓ Differentiate Mark-on, Mark-down, and Mark-up obtain Mark-on, Mark-down and Mark-up of a product (ABM_BM11BS-Ig-1)	**Worksheet for Week 5 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 5a. Explore. Activity 1. Can You Tell? #s 1-10 Module 5b, Explore. Activity 1. Complete Me! Letters – A, B, C	**March 21, 2022**
Quarter III Week 6 ***Key Concepts of Buying and Selling*** **March 21-25, 2022** ✓ Differentiate Mark-up from margins (ABM_BM11BS-Ih-3) ✓ Describe how gross margins is used in sales (ABM_BM11BS-Ih-4) ✓ Compute single trade discounts and discount series	**Worksheet for Week 6 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities Module 6. Gauge #s 1-10 Module 7. Gauge #s 1-10 Module 8. Enrichment. Activity 1. Solve Me #s 1-3	**March 28, 2022**

Quarter III Week *7 Key Concepts of Buying and Selling* **March 28- April 1, 2022** ✓ Differentiate profit from loss (ABM_BM11BS-Ii-6) ✓ Illustrate how profit is obtained and how to avoid loss in a given transaction (ABM_BM11BS-Ii-7)	**Worksheet for Week 7 (Reference Activity Sheet Quarter III)** Instructions: Answer the following activities Module 9. Explore #s 1-5 Module 10. Gauge #S 1-10	**April 4, 2022**
Quarter III Week *8 Key Concepts of Buying and Selling* **April 4-8, 2022** ✓ Determine the break-even point ✓ Solve problems involving buying and selling products (ABM_BM11BS-Ij-9)	**Worksheet for Week 8 (Reference: Activity Sheet Quarter 1II)** Instructions: Answer the following activities Module 11. Explore. Activity 1. Is it a Fact or a Bluff?	**April 11, 2022**
Quarter III Week 9 Key Concepts of Buying and Selling **April 11-13, 2022** ✓ Compute interest specifically as applied to mortgage, amortization and on services/ utilities and on deposits and loans	**Worksheet for Week 8 (Reference: Activity Sheet Quarter 1II)** Instructions: Answer the following activities Module 13. Gauge # 1-15	**April 18, 2022**